when atoms release these particles, they become smaller. This led him to propose that atoms disintegrate and that radiation is a by-product of the process. Rutherford also noticed that different radioactive materials disintegrate at different rates, which he called their "half-lives," since the time it takes for each material to reduce by half could be predicted.

Inside the atom

When he returned to England in 1907, Rutherford became professor of physics at the University of Manchester—and it was there that he made his most famous discovery. At the time, the accepted model of the atom was one proposed by Thomson, Rutherford's former mentor. Thomson imagined that an atom was a diffuse cloud of positive charge in which electrons were embedded. This was known as the "plum pudding" model, and to test it, two of Rutherford's students fired alpha particles at an ultra-thin film of gold foil. They expected the particles to pass through the gold atoms, albeit with slight deflections. However, while some of the particles passed through, others rebounded, suggesting that there was something in their way. Rutherford concluded that the mass and positive charge of an atom were all concentrated into a very small volume at its center, which he called its "nucleus."

Radical shift

In 1911, Rutherford published his model of the atom, which he likened to a miniature solar system: mostly empty space, with electrons orbiting a tiny nucleus, all held together by energy. This idea marked a profound leap, as it suggested that matter, at its most fundamental level, was not solid—a concept that was too radical for most of the scientific community, who liked the plum pudding model and thought that Rutherford's atom would not be able to remain stable.

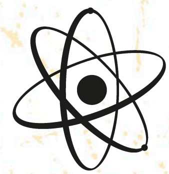
Rutherford succeeded Thomson as Cavendish Professor of Experimental Physics at Cambridge University in 1919, where he continued his research. He found that if nitrogen and other light elements are bombarded with alpha particles, they emit hydrogen nuclei. Hydrogen was known to be the simplest element, and Rutherford surmised that its nucleus must be one of the building blocks of all the elements. The hydrogen nucleus had a positive charge, so he named it the proton. He later proposed the existence of particles that have a neutral charge and reside within an atom's nucleus alongside the protons. The neutron (the name for such particles) was finally identified in 1932, by James Chadwick at the Cavendish Laboratory, under Rutherford's guidance.

NIELS **BOHR**

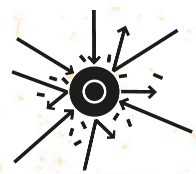
Along with Rutherford, Danish physicist Bohr played a key part in teasing out the structure of the atom.

Realizing that classical physics could not explain the Rutherford atomic model, Bohr (1885–1962) turned to quantum physics for the explanation. In 1913, he proposed that electrons have fixed levels of energy and orbit the nucleus at fixed distances in orbital "shells." Electrons change to higher or lower orbits by absorbing or emitting defined packets of energy (quanta).

DEMONSTRATED **THAT ATOMS** ARE NOT **INDESTRUCTIBLE**



DISCOVERED THE **ATOMIC NUCLEUS** AND THE **PROTON**



HAD A **SYNTHETIC ELEMENT** NAMED AFTER HIM: **RUTHERFORDIUM**